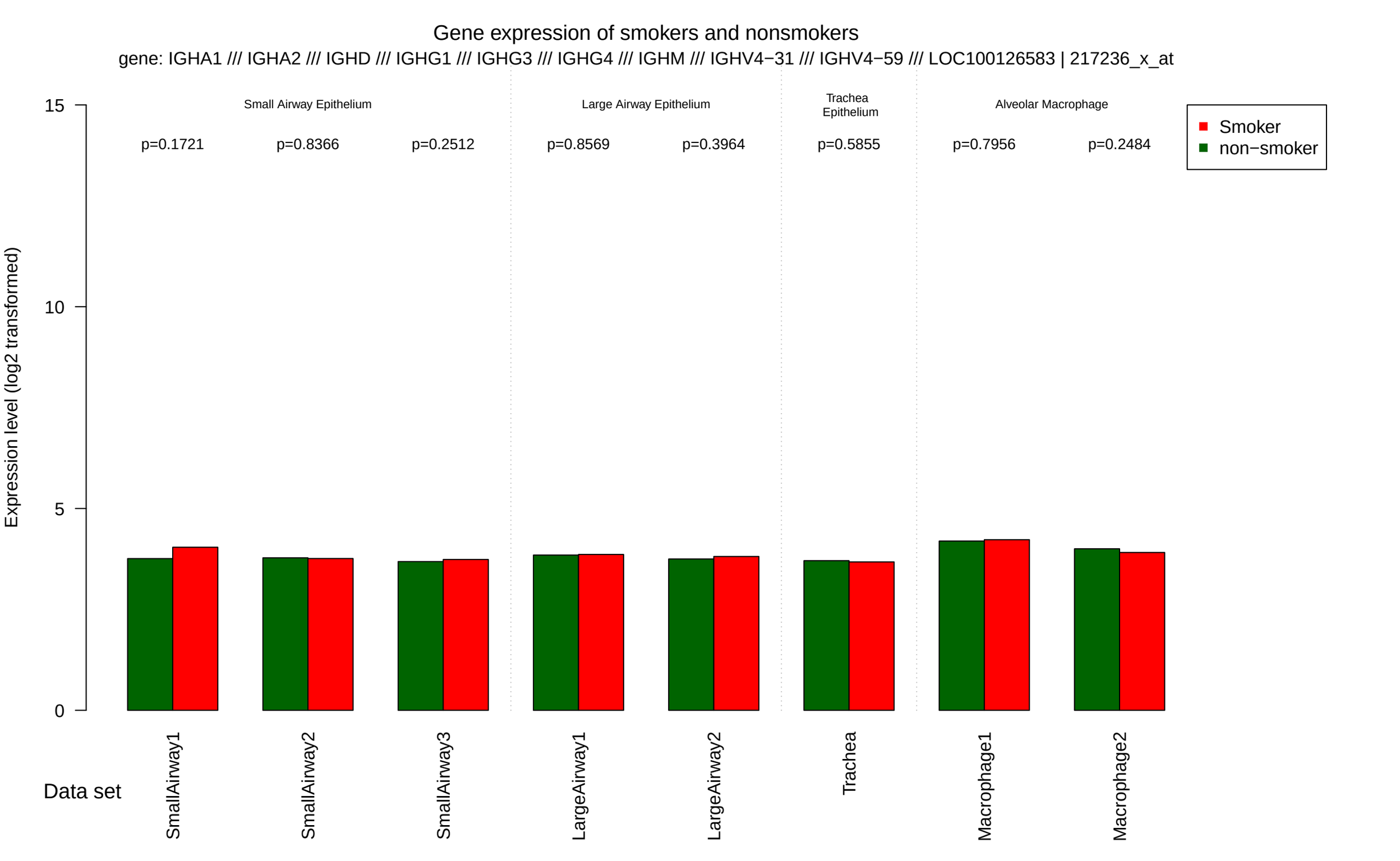
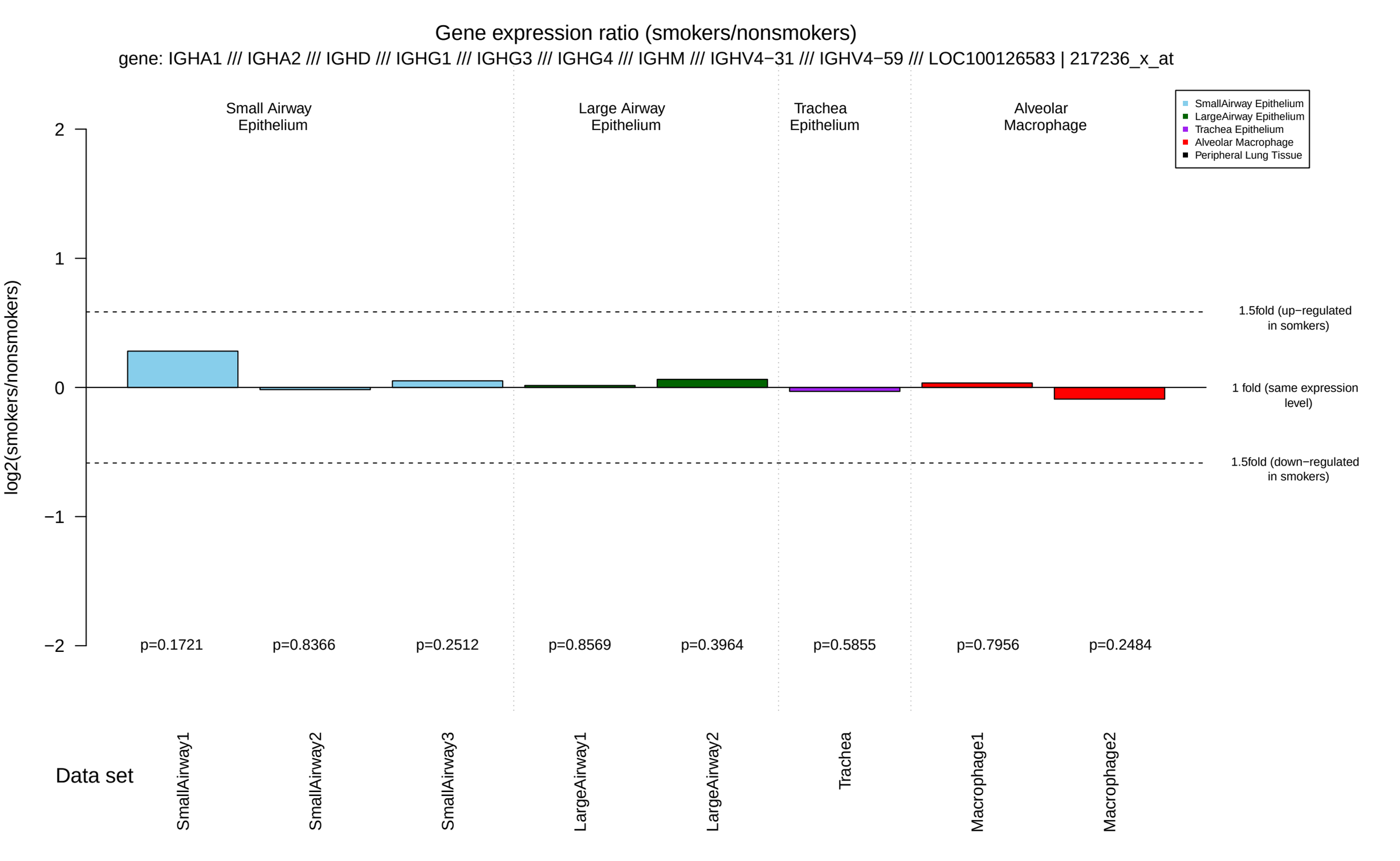
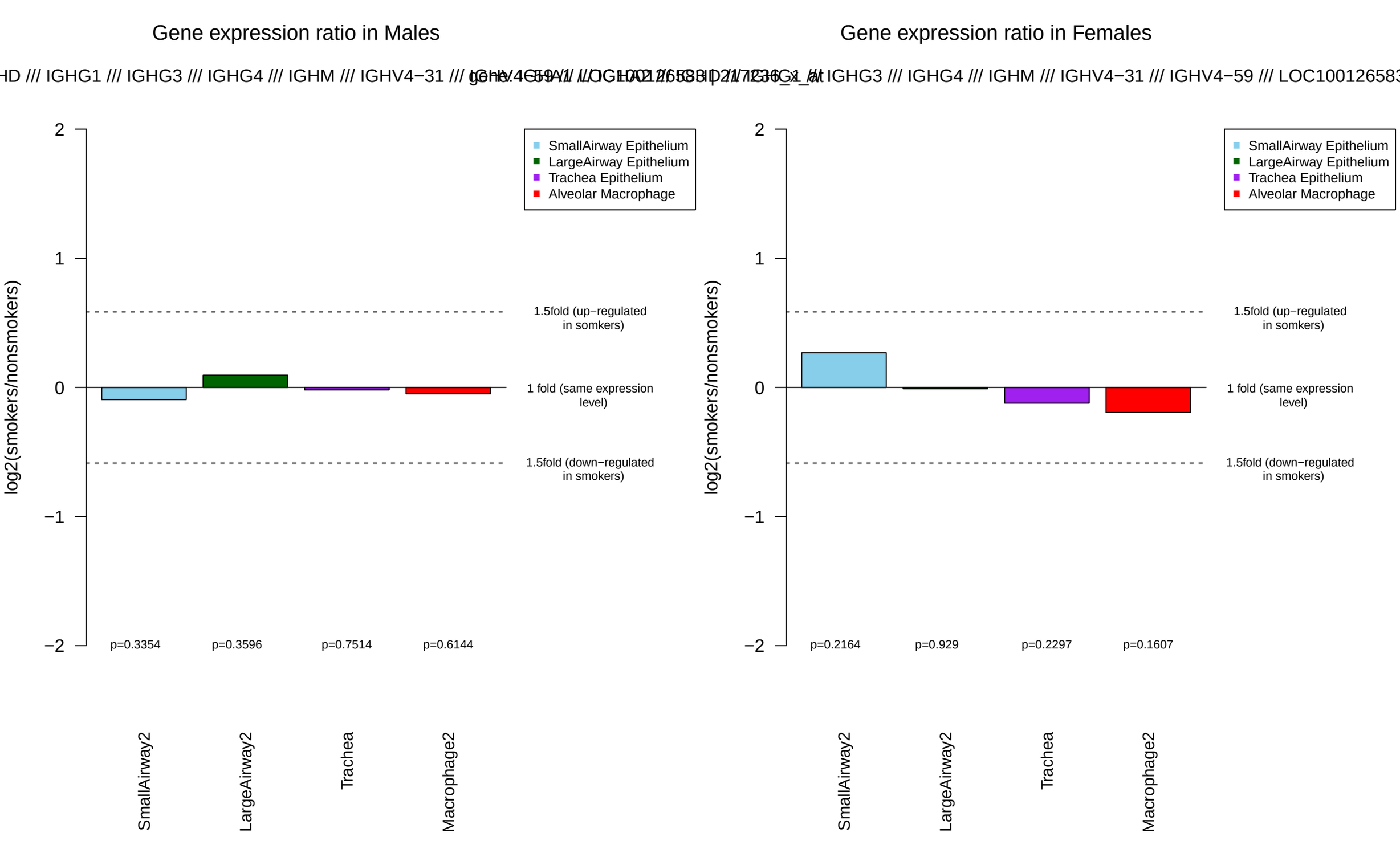
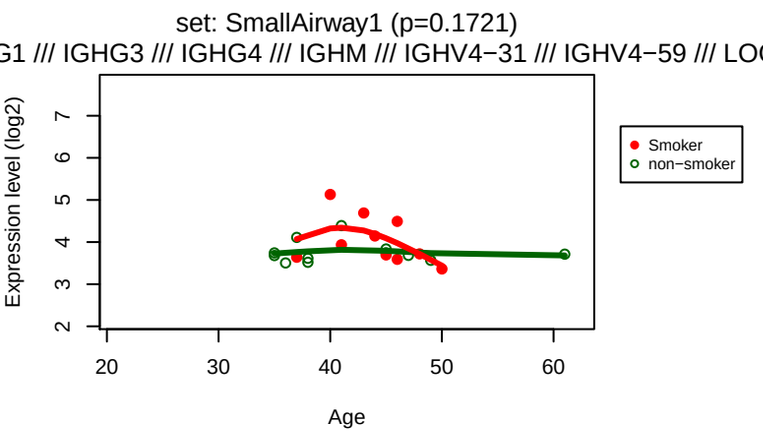
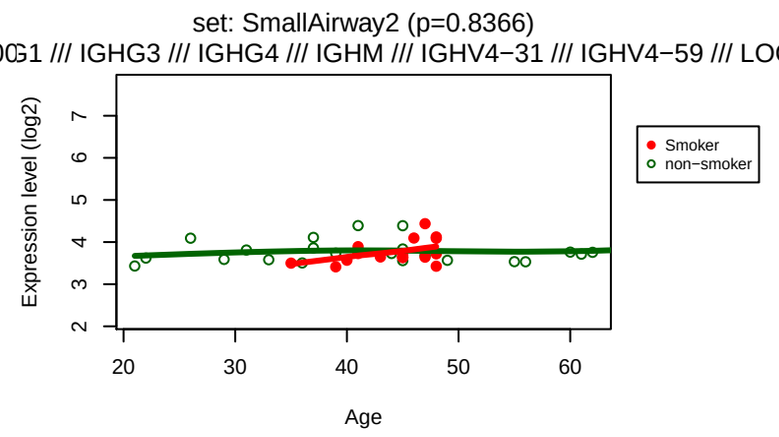
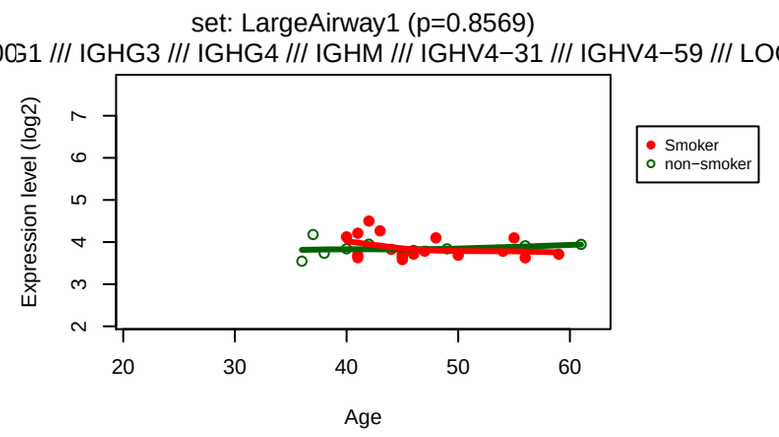

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 217236_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1721
p=0.8366
p=0.2512
p=0.8569
p=0.3964
p=0.5855
p=0.7956
p=0.2484
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

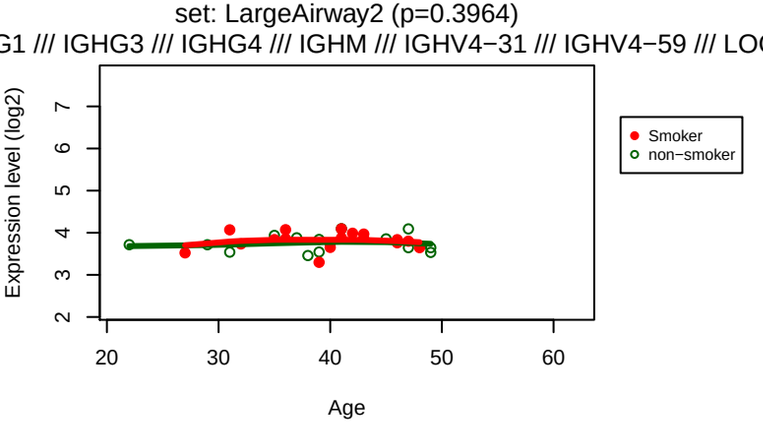

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 217236_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1721
p=0.8366
p=0.2512
p=0.8569
p=0.3964
p=0.5855
p=0.7956
p=0.2484
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

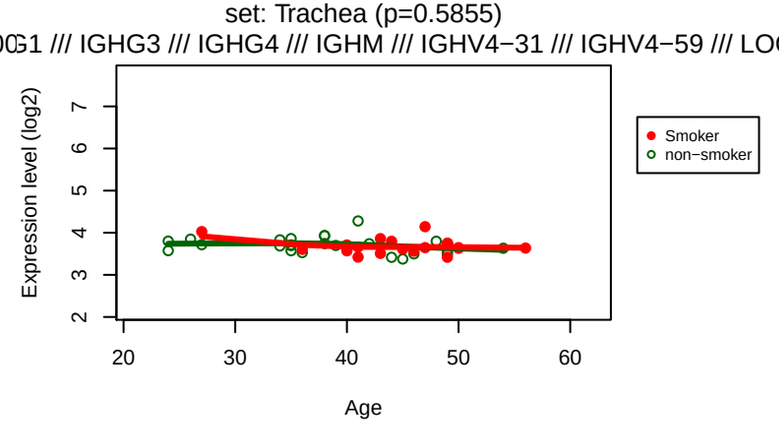

Gene expression ratio in Males
HD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 /// 217236_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3354
p=0.3596
p=0.7514
p=0.6144
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2164
p=0.929
p=0.2297
p=0.1607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

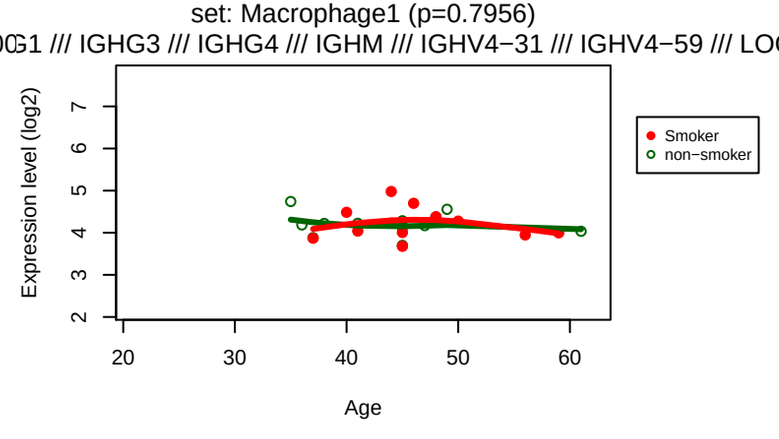

set: SmallAirway1 (p=0.1721)
G1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker

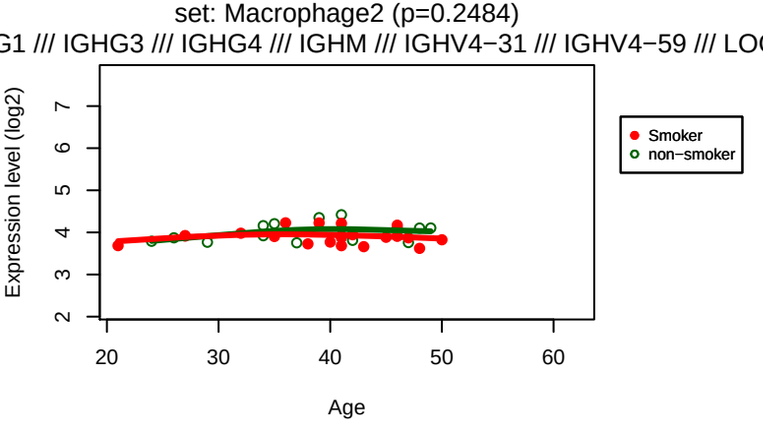

set: SmallAirway2 (p=0.8366)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker

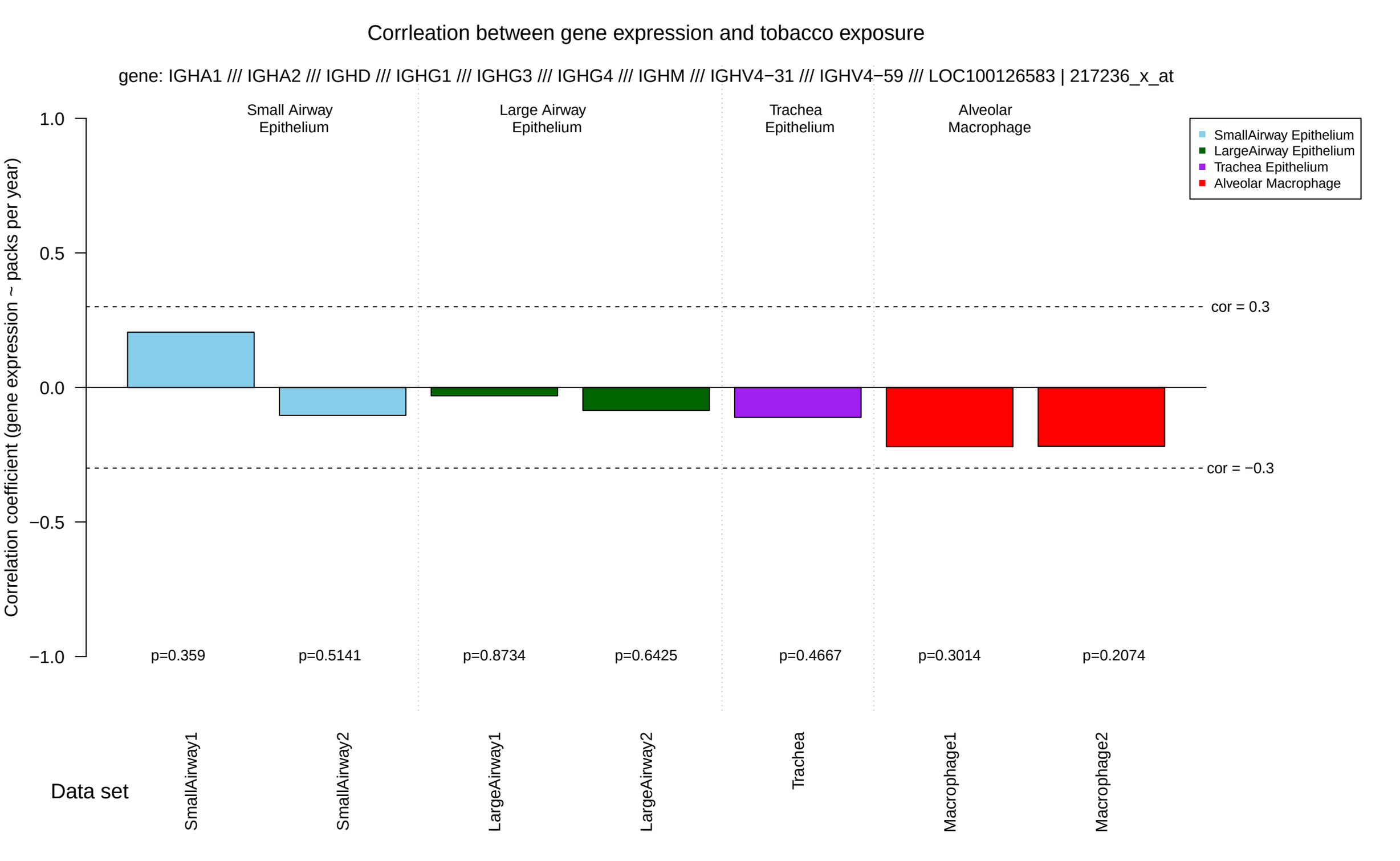

set: LargeAirway1 (p=0.8569)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3964)
G1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5855)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7956)
31 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2484)
G1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 217236_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.359
p=0.5141
p=0.8734
p=0.6425
p=0.4667
p=0.3014
p=0.2074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2